



Division of Mathematical Statistics

KTH Matematik

EXAM IN SF1930 STATISTICAL LEARNING AND DATA ANALYSIS
FRIDAY OCTOBER 22 2021, 08:00–13:00

Examiner: Joakim Andén, 08-790 80 52.

Allowed aids: Pocket calculator, formula sheet (without notes), statistical tables.

The exam consists of two parts, denoted part I and part II. Part I consists of problems 1–12. In this part, you only need to provide an answer, either in the form of a numerical value with at least *three significant digits* ("värdesiffror") or in the form of a choice between different possible answers. Students who passed the midterm exam ("kontrollskrivning") do not have to answer problems 1–3 and will receive full credit for them (this will be possible for the regular exam and the makeup exam of the same year only). The preliminary limit is 9 points for a passing grade (E) and 8 points for the possibility to submit a supplementary assignment to upgrade to a passing grade (Fx).

Part II consists of problems 13–16 and each correct solution gives 10 points. This part will only be graded for students who have passed part I or have passed the supplementary assignment to upgrade to a passing grade. The points on part II are used to obtain a grade higher than E. In this part, arguments and computations should be detailed and well-motivated so that they are easy to follow. Any notation you introduce must be explained and defined. Numerical answers must be provided with at least *three significant digits*. Students who passed the homework assignment will receive up to 5 bonus points on part II (again, this applies for the regular exam and the makeup exam of the same year only).

The exam will be graded within three working weeks from the exam date and will be available at the student office at least seven weeks after the exam date.

Part I

Problem 1

Let $\mathbf{X} = [X_1, X_2, X_3]^T \sim N(\boldsymbol{\mu}, \Sigma)$ where

$$\boldsymbol{\mu} = [1, -1, 0]^T \quad \text{and} \quad \Sigma = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

Calculate $\text{Var}[X_1 + 2X_2 + X_3]$.

Problem 2

Given a random sample $X_1, \dots, X_n$ from a gamma(α, β) (with the rate parametrization), where α and β are both unknown, which of the following is a sufficient statistic for (α, β) ?

- (a) $\prod_{i=1}^n X_i$,
- (b) $(\log(\sum_{i=1}^n X_i), \sum_{i=1}^n X_i)$,
- (c) $(\prod_{i=1}^n X_i, \sum_{i=1}^n X_i)$,
- (d) $(\sum_{i=1}^n \log(X_i), \sum_{i=1}^n X_i^{-1})$.

Problem 3

Let $X_1, X_2, \dots, X_n$ be an iid sample from a distribution such that

$$P(X_i = x) = \begin{cases} p, & x = -1 \\ 1 - p - q, & x = 0 \\ q, & x = 1 \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the method of moments estimator for (p, q) given the data 0, 1, 1, -1.

Problem 4

Suppose we are given data 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, 1 drawn from a bernoulli(p) distribution where $p \sim \text{beta}(4, 2)$. Assuming the samples are exchangeable, what is the posterior expectation of p given the data?

Problem 5

Suppose X has the pdf

$$f(x | \lambda) = \lambda e^{-\lambda x},$$

for $x \geq 0$ with some rate parameter $\lambda > 0$. We want to test $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda \neq \lambda_0$. Which of the following constitute an LRT for H_0 against H_1 (assuming that the constant c has been fixed to give the desired level α)?

- (a) Reject H_0 if $e^{-\lambda_0 x} \geq c$.
- (b) Reject H_0 if $\lambda_0 x e^{-\lambda_0 x} \geq c$.
- (c) Reject H_0 if $\lambda_0 e^{-\lambda_0 x} \leq c$.
- (d) Reject H_0 if $x e^{-\lambda_0 x} \leq c$.

Problem 6

A basketball player takes 2 free throws, making each shot, before finally missing on the third attempt. Assuming each shot was independent given the player's probability p of **missing** a single free throw, we can assume X , the first time a free throw is missed, is $\text{geom}(p)$. Based on past practice session data, we know that $p \sim \text{beta}(2, 2)$. Let a_0 denote the action "decide $0 \leq p < 0.45$," and let a_1 denote the action "decide $0.45 \leq p \leq 1$." Conduct a Bayes test with the given data. Give the posterior expected losses of the actions a_0 and a_1 and state which one you choose based on the test. The loss function is given by

$$L(p, a_0) = \begin{cases} 0 & \text{if } 0 \leq p < 0.45, \\ 1 & \text{if } 0.45 \leq p \leq 1, \end{cases} \quad L(p, a_1) = \begin{cases} 1 & \text{if } 0 \leq p < 0.45, \\ 0 & \text{if } 0.45 \leq p \leq 1. \end{cases}$$

Problem 7

Let X_1, X_2, X_3, X_4 be an iid sample from some distribution parameterized by θ . Suppose that for $H_0 : \theta = \theta_0$ versus $H_1 : \theta < \theta_0$, we have the level 10% test

$$\text{Reject } H_0 \text{ if } \bar{x} \leq \theta_0 - 0.82s,$$

where $\bar{x}$ and s^2 are the sample mean and sample variance, respectively. Compute a 90% one-sided confidence interval bounded from above given the data

$$1.93, \quad 2.38, \quad 1.63, \quad 3.53.$$

Problem 8

Suppose we have the following samples drawn from a $\text{Poisson}(\lambda)$ family:

$$6, 4, 6, 5, 7.$$

Assuming a $\text{gamma}(2, 4)$ prior on λ , give an equal-tailed 95% credible interval for λ .

Problem 9

Let $X_1, X_2, \dots$ be iid samples with the pmf

$$f(x) = \begin{cases} 1/4, & x = 0, \\ 1/2, & x = 1, \\ 1/4, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Set $Y_n = \frac{1}{n} \sum_{i=1}^n X_i$ for $n = 1, 2, \dots$. What is the asymptotic variance of $Y_1, Y_2, \dots$?

Problem 10

A factory that produces teddy bears counts, at the end of each day, the number of bears produced with defects (e.g., missing eye, odd number of arms, soulless gaze, etc...). The data from the last three days are given in the table below:

Day (i)	number of bears produced (n_i)	number of defective bears (x_i)
1	300	45
2	250	20
3	400	29

Letting θ_i denote the probability that a given bear produced on day i is defective, and assuming $\theta_1, \theta_2, \theta_3$ are a random sample from a $\text{beta}(\alpha, \beta)$ population, set up an empirical Bayes model and estimate the posterior mean $E[\theta_3 | x_3]$. When building your model, use maximum likelihood estimates to approximate the necessary θ_i 's and the method of moments to estimate α and β .

Problem 11

Suppose that $X \sim \text{binomial}(5, p)$, and we observe the data $X = 4$. Assuming a beta prior $p \sim \text{beta}(2, 5)$ compute the posterior probability that we observe the outcome $\tilde{X} = 1$.

Problem 12

Suppose that X has pdf

$$f(x) = \begin{cases} \frac{1}{x^2}, & x \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

How can we generate a sample X using a sample U from $\text{uniform}(0, 1)$?

Part II**Problem 13**

Pareto. A random variable X is said to be $\text{Pareto}(c, \alpha)$ for some $c, \alpha > 0$ if it has pdf

$$f(x) = \alpha c^\alpha x^{-(\alpha+1)}, \quad c \leq x < \infty.$$

Suppose that X and Y are two independent random variables where $X \sim \text{exponential}(1)$ and $Y \sim \text{gamma}(\alpha, 1)$ (rate parameterization). Find the distribution of $1 + \frac{X}{Y}$. (10 p)

Problem 14

The sad professor. A professor is conducting an oral exam for a course. Prior to the exam, she estimates that each student will pass with a probability of 0.8 on average, with a standard deviation of 0.1. This is done one student at a time, but she will stop the exam if 10 students fail the exam. (You may assume that the students are drawn with replacement.) For this course, she examines a total of 17 students before she is overwhelmed by sadness and stops the exam.

Once she recovers and decides to continue, what is the expected number of students that will pass in a class of 68 students in total? What is the standard deviation? (You may assume that this class of students is unrelated to the previous class except that it has the same probability of passing.) (10 p)

Problem 15

Models for joint binomial data. Suppose we have a sample of the form

$$\mathbf{X}^{(1)} = (1, 0), \quad \mathbf{X}^{(2)} = (0, 1), \quad \mathbf{X}^{(3)} = (1, 1), \quad \mathbf{X}^{(4)} = (0, 1), \quad \mathbf{X}^{(5)} = (0, 0),$$

from a joint distribution over (X_1, X_2) where X_1 and X_2 are binary random variables with possible outcomes 0 or 1. We can interpret our sample in two different ways: First, we assume that $(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \mathbf{X}^{(3)}, \mathbf{X}^{(4)}, \mathbf{X}^{(5)})$ is a size 5 sample drawn from a multinomial $(5, \theta_{(0,0)}, \theta_{(0,1)}, \theta_{(1,0)}, \theta_{(1,1)})$ population where $\theta_{(x_1, x_2)}$ denotes the probability of the outcome (x_1, x_2) . Alternatively, we can think of our sample as a sample of size 5 drawn from the joint distribution of two independent Bernoulli random variables $X_1 \sim \text{bernoulli}(p_1)$ and $X_2 \sim \text{bernoulli}(p_2)$.

- Show that the MLE of the independence model is $\hat{\theta}_{(x_1, x_2)} = \frac{n_{(x_1, x_2)}}{5}$, where $n_{(x_1, x_2)}$ denotes the number of $\mathbf{X}^{(k)} = (x_1, x_2)$ in our sample. (4 p)
- What is the MLE of (p_1, p_2) for the independence model? (2 p)
- Which of the two models is preferred according to their AIC scores given the data? (4 p)

Problem 16

Inference for the standard power distribution. Given $\beta > 0$, consider an iid sample $X_1, X_2, \dots, X_n$ with $n > 2$ and with pdf

$$f(x | \beta) = \begin{cases} \beta x^{\beta-1}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

- Calculate the MLE $\hat{\beta}$ for β given the sample. (2 p)
- Find a constant $c > 0$ such that $\beta^* = c\hat{\beta}$ is an unbiased estimator of β . *Hint:* To find the distribution of $\hat{\beta}$, take one transformation at a time. (5 p)
- Calculate the variance of β^* and compare this to the variance given by the Cramér–Rao lower bound. Is the estimator efficient? What happens as $n \rightarrow +\infty$? (3 p)

Good luck!



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SOLUTIONS TO EXAM IN SF1930 STATISTICAL LEARNING AND DATA ANALYSIS

Problem 1

Set $Y = X_1 + 2X_2 + X_3$, then $Y = A\mathbf{X}$ where $A = [1, 2, 1]$. Consequently, $Y \sim N(A\boldsymbol{\mu}, A\Sigma A^T)$. Now

$$A\Sigma A^T = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 31,$$

so $\text{Var}[Y] = 31$.

Problem 2

The likelihood function of the data can be written as

$$\begin{aligned} f(x_1, \dots, x_n \mid \alpha, \beta) &= \prod_{i=1}^n \frac{\beta^\alpha}{\Gamma(\alpha)} x_i^{\alpha-1} e^{-\beta x_i}, \\ &= \frac{\beta^{n\alpha}}{\Gamma(\alpha)^n} \left(\prod_{i=1}^n x_i \right)^{\alpha-1} e^{-\beta \sum_{i=1}^n x_i}. \end{aligned}$$

So the answer is (c).

Problem 3

From the pmf we have

$$\mathbb{E}[X_i] = -p + q \quad \text{and} \quad \mathbb{E}[X_i^2] = p + q.$$

Since

$$\frac{1}{4} \sum_{i=1}^4 x_i = \frac{1}{4} \quad \text{and} \quad \frac{1}{4} \sum_{i=1}^4 x_i^2 = \frac{3}{4},$$

we have to solve the following system of equations

$$\begin{cases} \frac{1}{4} = -p + q \\ \frac{3}{4} = p + q, \end{cases}$$

which has the solution $(p, q) = (1/4, 1/2)$.

Problem 4

Let $Y = \sum_{i=1}^{12} X_i$, where X_i denote our twelve bernoulli samples. Then $Y \sim \text{binomial}(12, p)$, and since $p \sim \text{beta}(4, 2)$, we know that $p \mid y \sim \text{beta}(11, 7)$. Hence, $\mathbb{E}[p \mid y] = 11/18$.

Problem 5

The likelihood function here is

$$L(\lambda | x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

To compute the LRT, we first need the (unconstrained) MLE, which for the case of one sample is given by $\hat{\lambda} = 1/x$. Plugging this in gives the maximum likelihood

$$L(\hat{\lambda} | x) = \frac{1}{x} e^{-\frac{1}{x}x} = \frac{e^{-1}}{x}.$$

The likelihood ratio is therefore given by

$$\ell(x) = \frac{\lambda_0 e^{-\lambda_0 x}}{e^{-1}/x} = e^1 \lambda_0 \cdot x e^{-\lambda_0 x}.$$

The LRT rejects H_0 if $\ell(x)$ is smaller than some constant. Equivalently, it rejects H_0 if $x e^{-\lambda_0 x} \leq c$ for some constant c . The answer is therefore (d).

Problem 6

Since $X \sim \text{geom}(p)$ and $p \sim \text{beta}(2, 2)$, it follows that for $x = 3$, $f(x | p) = p(1 - p)^2$ and $\pi(p) = \frac{\Gamma(4)}{\Gamma(2)\Gamma(2)} p(1 - p)$. Thus,

$$f(x) = \int_0^1 f(x, p) dp = \int_0^1 f(x | p) \pi(p) dp = \frac{\Gamma(4)}{\Gamma(2)\Gamma(2)} \int_0^1 p^2 (1 - p)^3 dp = \frac{\Gamma(4)}{\Gamma(2)\Gamma(2)} \frac{\Gamma(2)\Gamma(3)}{\Gamma(5)} = \frac{1}{2}.$$

Hence,

$$\pi(p | x) = \frac{f(x | p) \pi(p)}{f(x)} = \frac{\frac{\Gamma(4)}{\Gamma(2)\Gamma(2)} p^2 (1 - p)^3}{1/2} = 12 p^2 (1 - p)^3.$$

We want to pick the action a_0 or a_1 that minimizes the Bayes risk, or equivalently, that minimizes the posterior expected loss. For a_0 , the posterior expected loss is calculated as

$$\begin{aligned} \int_0^1 L(p, a_0) \pi(p | x) dp &= \int_{0.45}^1 1 \cdot (12 p^2 (1 - p)^3) dp, \\ &= 12 \int_{0.45}^1 (p^2 - 3p^3 + 3p^4 - p^5) dp \approx 0.088. \end{aligned}$$

For a_1 it is

$$\begin{aligned} \int_0^1 L(p, a_1) \pi(p | x) dp &= \int_0^{0.45} 1 \cdot (12 p^2 (1 - p)^3) dp, \\ &= 12 \int_0^{0.45} (p^2 - 3p^3 + 3p^4 - p^5) dp \approx 0.112. \end{aligned}$$

So we choose the action a_0 . (The ratio of the expected losses is $E[L(p, a_0)] / E[L(p, a_1)] \approx 0.786$. If you did not divide by the marginal probability $f(x) = 1/2$ in the above calculation you should have gotten $E[L(p, a_0)] \approx 0.044$ and $E[L(p, a_1)] \approx 0.055$. We also accepted these answers.)

Problem 7

Since the test rejects $H_0 : \theta = \theta_0$ if $\bar{x} \leq \theta_0 - 0.82s$, it *accepts* H_0 if $\bar{x} \geq \theta_0 - 0.82s$. Per definition, the probability of acceptance when H_0 is true is at least 0.90. We thus have

$$P(\bar{X} \geq \theta_0 - 0.82S \mid \theta_0) \geq 0.9$$

Reshuffling this, we obtain

$$P(\theta_0 \leq \bar{X} + 0.82S \mid \theta_0) \geq 0.9.$$

In other words, an upper-bounded 90% confidence interval is given by $(-\infty, \bar{x} + 0.82s] \approx (-\infty, 3.05]$, where we have plugged in $\bar{x} \approx 2.367$ and $s \approx 0.8340$.

Problem 8

Since $X \sim \text{Poisson}(\lambda)$ and $\lambda \sim \text{gamma}(2, 4)$, for data $\mathbf{x} = (x_1, \dots, x_n)$ we have that $\lambda \mid \mathbf{x} \sim \text{gamma}(\alpha + \sum_i x_i, \beta + n)$. So in our case, $\lambda \mid \mathbf{x} \sim \text{gamma}(30, 9)$. Recall that if $Y \sim \text{gamma}(\alpha, \beta)$ then

$$2\beta Y \sim \text{gamma}(\alpha, 1/2) = \chi^2(2\alpha).$$

Hence, $2 \cdot 9(\lambda \mid \mathbf{x}) \sim \chi^2(60)$ in our case. Using the table for a χ^2 distribution we calculate that an approximate 95% credible interval for λ given the data is

$$\left[\frac{1}{18} \chi_{1-2.5}^2(60), \frac{1}{18} \chi_{2.5}^2(60) \right] = \left[\frac{1}{18} (40.5), \frac{1}{18} (83.3) \right] = [2.25, 4.63].$$

Problem 9

By the central limit theorem,

$$\sqrt{n}(Y_n - \mu) \longrightarrow N(0, \sigma^2)$$

in distribution, where $\mu = E[X]$ and $\sigma^2 = \text{Var}[X]$, so the asymptotic variance is $\text{Var}[X]$. In this case, we compute

$$E[X] = \frac{1}{4} \cdot 0 + \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 = 1$$

and

$$\text{Var}[X] = \frac{1}{4} \cdot 1 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 1 = \frac{1}{2},$$

so the asymptotic variance is $1/2$.

Problem 10

We estimate $\theta_1 \approx \frac{45}{300}$ and $\theta_2 \approx \frac{20}{200}$, and then take the first and second sample moments

$$m_1 = 0.115, \quad m_2 = 0.016.$$

Using the first and second moments for beta distributions as given on the formula sheet, we then solve the system of equations:

$$m_1 = \frac{\alpha}{\alpha + \beta}, \quad m_2 = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)},$$

which has solution given by

$$\alpha = m_1 \left(\frac{m_2 - m_1}{m_1^2 - m_2} \right), \quad \beta = \alpha \left(\frac{1 - m_1}{m_1} \right).$$

Thus, we estimate $\alpha = 4.103$ and $\beta = 31.575$. Since the beta family is conjugate to the binomial family we see that

$$\pi(\theta_3 | x_3) \propto \theta_3^{33.103} (1 - \theta_3)^{402.575},$$

which means that the posterior mean is

$$E[\theta_3 | x_3] = \frac{32.103}{32.103 + 401.575} \approx 0.074.$$

(So the probability of a teddy bear having soulless eyes is quite small... fortunately.)

If we alternatively solved a system of equations based on the sample mean and variance we would get a different estimate for α and β but essentially the same answer for the posterior mean:

$$\begin{aligned} \bar{\theta} &= \frac{1}{2} \left(\frac{45}{300} + \frac{20}{200} \right) \approx 0.115, \\ s^2 &= \left(\frac{45}{300} - \bar{\theta} \right)^2 + \left(\frac{20}{200} - \bar{\theta} \right)^2 \approx 0.001. \end{aligned}$$

Since the θ_i 's are samples from a $\text{beta}(\alpha, \beta)$ population, we solve the system of equations

$$0.115 = \frac{\alpha}{\alpha + \beta}, \quad 0.001 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)},$$

which yields

$$\alpha = \bar{\theta} \left(\frac{\bar{\theta}(1 - \bar{\theta})}{s^2} - 1 \right), \quad \beta = \alpha \frac{(1 - \bar{\theta})}{\bar{\theta}}.$$

So we estimate $\alpha \approx 9.4$ and $\beta \approx 72.6$. Since the beta family is conjugate to the binomial family we see that

$$\pi(\theta_3 | x_3) \propto \theta_3^{32.4} (1 - \theta_3)^{442.6},$$

which means that the posterior mean is

$$E[\theta_3 | x_3] = \frac{32.4}{32.4 + 442.6} \approx 0.07.$$

Problem 11

Since $X \sim \text{binomial}(5, p)$ and $p \sim \text{beta}(2, 5)$, we know that $f(x | p) = \binom{5}{x} p^x (1 - p)^{5-x}$ and $\pi(p) = \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)} p(1 - p)^4$. It follows that for $x = 4$,

$$\pi(p | x) = \frac{\Gamma(12)}{\Gamma(6)\Gamma(6)} p^5 (1 - p)^5.$$

We are interested in the posterior probability of the outcome $\tilde{x} = 1$. This is computed as

$$\begin{aligned}
 f(\tilde{x} \mid x) &= \int_0^1 f(\tilde{x} \mid p) \pi(p \mid x) dp, \\
 &= \int_0^1 \binom{5}{1} p(1-p)^4 \frac{\Gamma(12)}{\Gamma(6)\Gamma(6)} p^5 (1-p)^5 dp, \\
 &= 5 \frac{\Gamma(12)}{\Gamma(6)\Gamma(6)} \int_0^1 p^6 (1-p)^9 dp, \\
 &= 5 \frac{\Gamma(12)}{\Gamma(6)\Gamma(6)} \frac{\Gamma(7)\Gamma(10)}{\Gamma(17)}, \\
 &\approx 0.173.
 \end{aligned}$$

Problem 12

The cdf is given by

$$F(x) = \int_1^x f(t) dt = \int_1^x \frac{1}{t^2} dt = \left[-\frac{1}{t} \right]_{t=1}^x = 1 - \frac{1}{x}.$$

Setting $F(x) = u$ and inverting gives $x = 1/(1-u)$ so we can generate a sample from X by computing $1/(1-U)$ or just $1/U$ (since U and $1-U$ have the same distribution).

Problem 13

Letting $U = X$ and $V = 1 + \frac{X}{Y}$, we find that $X = U$ and $Y = \frac{U}{V-1}$. Hence, the Jacobian of the inverse transformation is

$$M = \left| \det \begin{pmatrix} 1 & 0 \\ \frac{1}{v-1} & \frac{-u}{(v-1)^2} \end{pmatrix} \right| = \frac{u}{(v-1)^2}.$$

It follows that the joint pdf of (U, V) is

$$\begin{aligned}
 f_{U,V}(u, v) &= f_{X,Y}(u, \frac{u}{v-1}) \frac{u}{(v-1)^2}, \\
 &= \frac{1}{\Gamma(\alpha)} \left(\frac{u}{v-1} \right)^{\alpha-1} e^{-u/(v-1)} \frac{u}{(v-1)^2}.
 \end{aligned}$$

To compute the pdf of V we marginalize out $U = X$, which has domain $0 < u < \infty$:

$$\begin{aligned}
 f_V(v) &= \int_0^\infty \frac{1}{\Gamma(\alpha)} \frac{u^\alpha}{(v-1)^{\alpha+1}} e^{-u/(v-1)} du, \\
 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} \frac{\left(\frac{w}{(v/(v-1))} \right)^\alpha}{(v-1)^{\alpha+1}} e^{-w} \frac{v-1}{v} dw, \quad w = u(v/(v-1)), \\
 &= \int_0^\infty \frac{1}{\Gamma(\alpha)} v^{-(\alpha+1)} w^\alpha e^{-w} dw, \\
 &= \frac{1}{\Gamma(\alpha)} v^{-(\alpha+1)} \int_0^\infty w^\alpha e^{-w} dw, \\
 &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} v^{-(\alpha+1)}, \\
 &= \alpha v^{-(\alpha+1)}.
 \end{aligned}$$

Noting that if $V < 1$ the $f_V(v) = 0$, we conclude that $1 + \frac{X}{Y} \sim \text{Pareto}(1, \alpha)$.

Problem 14

The natural model for this problem is to take the probability of success P to be a $\text{beta}(\alpha, \beta)$ random variable. Given that we want $E[P] = \mu = 0.8$ and $\text{Var}[P] = \sigma^2 = 0.1^2 = 0.01$, this gives the system of equations

$$\frac{\alpha}{\alpha + \beta} = \mu \quad \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \sigma^2.$$

Using the fact that $1 - \mu = \beta/(\alpha + \beta)$, we have

$$\mu(1 - \mu) = \frac{\alpha\beta}{(\alpha + \beta)^2} = \sigma^2(\alpha + \beta + 1),$$

so

$$\alpha + \beta = \frac{\mu(1 - \mu)}{\sigma^2} - 1.$$

Consequently, we have

$$\alpha = \left(\frac{\mu(1 - \mu)}{\sigma^2} - 1 \right) \mu \quad \beta = \left(\frac{\mu(1 - \mu)}{\sigma^2} - 1 \right) (1 - \mu),$$

which gives, in our case, $\alpha = 12$ and $\beta = 3$.

Since she stops after 10 students fail and she has examined a total of 17 students, that means that 7 students passed. The number of students who pass, conditioned on P , is distributed as $\text{binomial}(n, P)$, where n is the total number of students examined. The posterior is then given by

$$P \mid x \sim \text{beta}(\alpha + x, \beta + n - x),$$

which gives, in our case, $\text{beta}(12 + 7, 3 + 10) = \text{beta}(19, 13)$.

We are interested in $Y \mid p \sim \text{binomial}(n, p)$, which means that Y has a beta-binomial distribution. We could calculate the pmf of Y and use this to compute mean and standard deviation, but the tower rules simplify things here. For the mean, we have

$$E[Y] = E[E[Y \mid P]] = E[nP] = \frac{n\alpha}{\alpha + \beta},$$

while the variance is given by

$$\text{Var}[Y] = \text{Var}[E[Y \mid P]] + E[\text{Var}[Y \mid P]] = \text{Var}[nP] + E[nP(1 - P)] = n^2 \text{Var}[P] + n E[P] - n E[P^2].$$

Using the fact that $E[P^2] = \text{Var}[P] + E[P]^2$ and regrouping, we get

$$\text{Var}[Y] = n^2 \text{Var}[P] + n E[P] - n \text{Var}[P] - n E[P]^2 = n(n - 1) \text{Var}[P] + n E[P](1 - E[P]).$$

Now plugging in $E[P] = \alpha/(\alpha + \beta)$ and $\text{Var}[P] = \alpha\beta/(\alpha + \beta)^2(\alpha + \beta + 1)$, we obtain

$$\text{Var}[Y] = \frac{n\alpha\beta(\alpha + \beta + n)}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

From our data, we have $\alpha = 19$, $\beta = 13$ and $n = 68$, which gives a mean of 40.375 and standard deviation of about $\sqrt{49.7041} \approx 7.05$.

Problem 15

- (a) The log-likelihood of the multinomial model is

$$\log f(x_{(i,j)} \mid 5, \theta_{(i,j)}) = \log 60 \cdot \theta_{(0,0)}^{x_{(0,0)}} \theta_{(0,1)}^{x_{(0,1)}} \theta_{(1,0)}^{x_{(1,0)}} \theta_{(1,1)}^{x_{(1,1)}},$$

where $\sum_{(i,j)} x_{(i,j)} = 5$ and $\sum_{(i,j)} \theta_{(i,j)} = 1$. To compute the MLE of the model we consider the Lagrangian for the constraint $\sum_{(i,j)} \theta_{(i,j)} = 1$

$$g(\theta_{(i,j)}, \lambda) = \log f(x_{(i,j)} \mid 5, \theta_{(i,j)}) - \lambda \left(1 - \sum_{(i,j)} \theta_{(i,j)} \right).$$

Differentiating with respect to $\theta_{(i,j)}$ yields

$$\frac{\partial}{\partial \theta_{(i,j)}} g(\theta_{(i,j)}, \lambda) = \frac{x_{(i,j)}}{\theta_{(i,j)}} - \lambda.$$

Setting these equations equal to zero and solving for $\theta_{(i,j)}$ yields

$$\theta_{(i,j)} = \frac{x_{(i,j)}}{\lambda}.$$

However, we also know that $\sum_{(i,j)} x_{(i,j)} = 5$. So since

$$1 = \sum_{(i,j)} \theta_{(i,j)} = \sum_{(i,j)} \frac{x_{(i,j)}}{\lambda},$$

we conclude that $\hat{\theta}_{(i,j)} = \frac{x_{(i,j)}}{5}$ since the log-likelihood is a concave function on its domain.

- (b) Let $p_1 = f_{X_1}(1)$ and $p_2 = f_{X_2}(1)$. It follows that the log-likelihood of the data is

$$\log f(x_1, x_2 \mid p_1, p_2) = \log p_1^2 (1 - p_1)^3 p_2^3 (1 - p_2)^2.$$

Computing the partial derivative with respect to p_1 and setting it equal to zero yields

$$\begin{aligned} \frac{2}{p_1} &= \frac{3}{1 - p_1}, \\ p_1 &= \frac{2}{5}, \end{aligned}$$

and similarly $p_2 = 3/5$. Since we have one likelihood equation for each parameter, and these are the likelihood equations for their respective (univariate) models, then we can apply the fact that these are the global maximizers for each variable to deduce that the MLE is $(\hat{p}_1, \hat{p}_2) = (2/5, 3/5)$. (Note that this is just applying independence of the variables X_1, X_2 .)

- (c) To compute the AIC of the multinomial model we evaluate the deviance at the MLE and add two times the number of model parameters. Since $\sum_{(i,j)} \theta_{(i,j)} = 1$, this number is three. Hence

$$\text{AIC} = -2 \log \left(60 \left(\frac{1}{5} \right) \left(\frac{2}{5} \right)^2 \left(\frac{1}{5} \right) \left(\frac{1}{5} \right) \right) + 2(3) \approx 7.914.$$

The independence model on the other hand has two free parameters, p_1, p_2 . Hence, its AIC is

$$\text{AIC} = -2 \log \left(\binom{5}{2} \binom{5}{3} \left(\frac{2}{5} \right)^2 \left(\frac{3}{5} \right)^3 \left(\frac{3}{5} \right)^3 \left(\frac{2}{5} \right)^2 \right) + 2(2) \approx 9.061.$$

Hence, the multinomial model has a lower AIC and is therefore the preferred model.

Problem 16

(a) We form the log-likelihood

$$\begin{aligned} \log L(\beta \mid x_1, x_2, \dots, x_n) &= \log (\beta^n (x_1 x_2 \cdots x_n)^{\beta-1}) \\ &= n \log \beta + (\beta - 1) \sum_{i=1}^n \log x_i. \end{aligned}$$

Taking the derivative, we have

$$\frac{d}{d\beta} \log L(\beta \mid x_1, x_2, \dots, x_n) = \frac{n}{\beta} + \sum_{i=1}^n \log x_i$$

and setting this to zero gives

$$\hat{\beta} = -\frac{n}{\sum_{i=1}^n \log x_i}.$$

Since the second derivative

$$\frac{d^2}{d\beta^2} \log L(\beta \mid x_1, x_2, \dots, x_n) = -\frac{n}{\beta^2}$$

is always negative, the likelihood is a concave function and its one critical point must therefore be a global maximum.

(b) We want to compute the distribution of

$$-\frac{n}{\sum_{i=1}^n \log X_i} = \frac{n}{\sum_{i=1}^n -\log X_i}$$

First, if we set $Y = -\log X$, then we have the change of variable $y = -\log x$ where $y \in (0, +\infty)$ with inverse $x = e^{-y}$ and Jacobian $|df/dx| = e^{-y}$. This gives

$$f_Y(y) = f_X(e^{-y})e^{-y} = \beta e^{-y(\beta-1)}e^{-y} = \beta e^{-\beta y},$$

so $Y \sim \text{exponential}(\beta)$ in the rate parameterization.

So we have that $-\log X_i \sim \text{exponential}(\beta)$ and we want to compute the distribution of $\sum_{i=1}^n -\log X_i$. Using change of variables or the convolution formula (from the previous course), we can show that, if $U \sim \text{gamma}(k, \beta)$ and $V \sim \text{exponential}(\beta)$, we have that $T = U + V$ has the pdf

$$\begin{aligned} f(t) &= \int_0^t \frac{\beta^k}{\Gamma(k)} v^{k-1} e^{-\beta v} \cdot \beta e^{-\beta(t-v)} dv = \frac{\beta^{k+1}}{\Gamma(k)} e^{-\beta t} \int_0^t v^{k-1} dv \\ &= \frac{\beta^{k+1}}{\Gamma(k)} e^{-\beta t} \frac{t^k}{k} = \frac{\beta^{k+1}}{\Gamma(k+1)} t^k e^{-\beta t}. \end{aligned}$$

In other words, $T \sim \text{gamma}(k+1, \beta)$.

Iterating this, we have that $-\sum_{i=1}^n \log X_i \sim \text{gamma}(n, \beta)$ since each term $-\log X_i \sim \text{exponential}(\beta) = \text{gamma}(1, \beta)$, which means that

$$-\frac{1}{\sum_{i=1}^n \log X_i}$$

has an inverse-gamma($n, 1/\beta$) distribution (parameterized by rate), which gives

$$\mathbb{E}[\hat{\beta} \mid \beta] = \mathbb{E}\left[-\frac{n}{\sum_{i=1}^n \log X_i} \mid \beta\right] = n \mathbb{E}\left[-\frac{1}{\sum_{i=1}^n \log X_i} \mid \beta\right] = n \frac{\beta}{n-1} = \frac{n}{n-1} \beta$$

using the formula for the expectation of the inverse-gamma distribution in the formula sheet. Consequently, the desired unbiased estimator is given by

$$\beta^* = \frac{n-1}{n} \hat{\beta} = -\frac{n-1}{\sum_{i=1}^n \log X_i}.$$

(c) Again, using the formulas from the formula sheet, we have

$$\begin{aligned} \text{Var}[\beta^* \mid \beta] &= \text{Var}\left[\frac{n-1}{n} \hat{\beta} \mid \beta\right] = \left(\frac{n-1}{n}\right)^2 \text{Var}[\hat{\beta} \mid \beta] \\ &= \left(\frac{n-1}{n}\right)^2 \cdot n^2 \cdot \frac{\beta^2}{(n-1)^2(n-2)} = \frac{\beta^2}{n-2}. \end{aligned}$$

The CRLB, on the other hand, is calculated from the information number

$$I(\beta) = -\mathbb{E}\left[\frac{d^2}{d\beta^2} \log L(\beta \mid X)\right] = -\mathbb{E}\left[-\frac{n}{\beta^2}\right] = \frac{n}{\beta^2},$$

which gives the bound

$$\text{Var}[\beta^* \mid \beta] \geq \frac{1}{I(\beta)} = \frac{\beta^2}{n}.$$

Clearly β^* does not achieve this bound and the estimator is therefore not efficient. As $n \rightarrow +\infty$, however, the ratio between the variances goes to one and the estimator is therefore asymptotically efficient (which is expected given the asymptotic efficiency properties of the MLE).